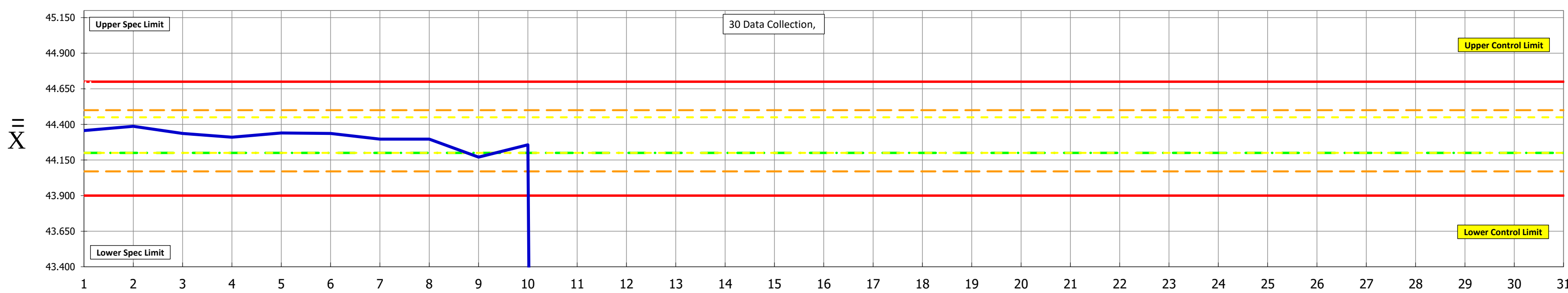
	PT.SUMITOMO ELECTRIC WINTEC INDONESIA	No. Dokumen: 4-QAS-PP07-001
	$\overline{X}$ - R CONTROL CHART	Revisi: 00
		Tgl Berlaku: 2-Dec-16

MACHINE	SECTION	OPERATION	DATE CONTROL LIMITS CALC.	SPECIFICATION	COURSE NO.	CONTROL ITEM (Δ) YES ○
CONDUCTOR RESISTANCE	**QC**	**Conductor Resistance**		min max **45.45**		NO
	MONTH **Jul-25**	CHARACTERISTIC **Resistance (Hambatan)**	SAMPLE SIZE / FREQUENCY n = **3**		TYPE / SIZE : **D2IDW4-B-2 / 0.70** PIC : **NISA AMALIAH**	

$\overline{X}$ = AVERAGE $\overline{X}$ = **44.200** U.C.L. = $\overline{\overline{X}} + A_2\overline{R}$ = **44.7000** L.C.L. = $\overline{\overline{X}} - A_2\overline{R}$ = **43.9000**

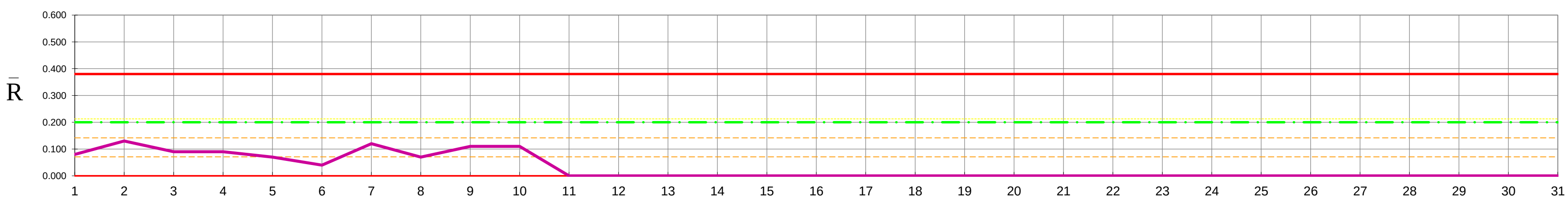
AVERAGE (X BAR CHART)



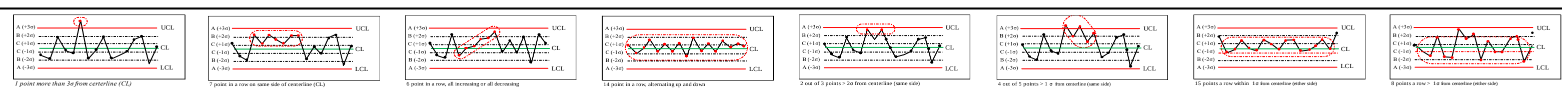
$\overline{R}$ = AVERAGE R = **0.200** U.C.L. = $D_4\overline{R}$ = **0.380** L.C.L. = $D_3\overline{R}$ = **0.00**

Special Cause Marker Legend:
Test 1 ● Test 2 ◆ Test 3 ▲ Test 4 ■

RANGES (R - CHART)



Special Cause



Month		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
DATA	1	44.350	44.450	44.390	44.360	44.380	44.330	44.230	44.260	44.160	44.310						
	2	44.320	44.320	44.320	44.300	44.330	44.360	44.350	44.300	44.230	44.260						
	3	44.400	44.390	44.300	44.270	44.310	44.320	44.310	44.330	44.120	44.200						
N		3	3	3	3	3	3	3	3	3	3	0	0	0	0	0	0
AVERAGE, X		44.357	44.387	44.337	44.310	44.340	44.337	44.297	44.297	44.170	44.257	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Range		0.080	0.130	0.090	0.090	0.070	0.040	0.120	0.070	0.110	0.110						
Moving Range																	
SHIFT 1		AHS	NISA		ANDI	NISA	NISA	R	R				sun		R		
SHIFT 2																	

Date	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AVERAGE, X	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Range															
Moving Range															
SHIFT 1		SUN						sun							
SHIFT 2															

SPECIAL CAUSES

- ☑ TEST 1: ANY POINT OUT OF CONTROL LIMIT
- ☑ TEST 2: 7 POINTS IN A ROW ON SAME SIZE OF CENTER LINE
- ☑ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
- ☑ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
- ☑ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
- ☑ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
- ☐ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
- ☑ TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

ACTION INSTRUCTION

1. ______
2. ______
3. ______
4. ______
5. ______

SUBGROUP SIZE	A_2	d_2	D_3	D_4
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

CALCULATION

$\overline{X}$ = 44.3087
$\overline{R}$ = 0.0910
s_{ST} = 0.0538 s_{LT} = 0.071

THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED

$$C_{PK} = \text{MIN}\left[\frac{\overline{X} - LSL}{3s_{ST}} \quad \frac{USL - \overline{X}}{3s_{ST}}\right] = \textbf{MIN}\left[\ 2.53 \quad 2.43\ \right] = 2.43$$

$$C_p = \left[\frac{USL - LSL}{6s_{ST}}\right] = \left[\ 2.48\ \right]$$

CHECKED BY	APPROVED BY